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Memory based cuckoo search algorithm for feature selection of gene expression dataset

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ABSTRACT

Cancer prediction has been shown to be important in the cancer research area. This importance has prompted many researchers to review machine learning-approaches to predict cancer outcome using gene expression dataset. This dataset consists of many genes (features) which can mislead the prediction ability of the machine learning methods, as some features may lead to confusion or inaccurate classification. Since finding the most informative genes for cancer prediction is challenging, feature selection techniques are recommended to pick important and relevant features out of large and complex datasets. In this research, we propose the Cuckoo search method as a feature selection algorithm, guided by the memory-based mechanism to save the most informative features that are identified by the best solutions. The purpose of the memory is to keep track of the selected features at every iteration and find the features that enhance classification accuracy. The suggested algorithm has been contrasted with the original algorithm using microarray datasets and the proposed algorithm has been shown to produce good results as compared to original and contemporary algorithms.

1. Introduction

Recently, feature selection, has become an attractive cancer research field. Feature selection is recognized as an NP-hard issue [1,2]. The complexity of the problem arises in selecting the most informative features that will help the prediction methods to classify with high accuracy the data with a minimum number of features and a satisfactory performance.

Cancer is a disease that occurs when one or more cells start to undergo mutation. This can happen during cell growth when the cell starts to react in abnormal ways, such as replicating itself uncontrollably. With a series of mutations, the cancer cell might spread to other body parts and thus infect other cells as well.

Today, cancer classification has used advanced techniques such as microarray technology to conduct research. Microarray data can enable measuring thousands of genes simultaneously, with the genes expression dataset as an output [3]. This technology also has successfully been applied in many problems and has achieved a superior result compared with other techniques, particularly in the medical field. Microarray also has shown the ability to diagnose patients who have a specific disease. Thus, this technology is used to detect diseases such as cancer. The most important weakness of microarray dataset is the large dimensionality and the complex interrelationship between features [4]. To solve these issues irrelevant genes should be eliminated and the dimensionality of

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Table 1
Dataset descriptions.

Dataset	Samples	Genes	Classes	Disease
Alon	62	2000	2	Colon Cancer [14]
Golub	72	7129	2	Leukemia [15]
Gordon	181	12,533	2	Lung Cancer [16]
Gravier	168	2905	2	Breast Cancer [17]
Pomeroy	60	7128	2	CNSET [18]
Shipp	58	6817	2	Lymphoma [19]
Subramanian	50	10,100	2	N/A [20]
Tian	173	12,625	2	Myeloma [21]
West	49	7129	2	Breast Cancer [22]
Chiaretti	128	12,625	2	Leukemia [23]
Sun	180	54,613	4	Glioma [24]
Yeoh	248	12,625	6	Leukemia [25]

the microarray data also should be reduced [5,6].

In gene expression, the most common task is a feature or gene selection where usually the aim is to identify the best gene that has a high predictive performance. In classification procedure it is important to categorize data into two or many classes that have the same characteristics. In the biomedical field, the most difficult part is classification when using microarray data that have high dimensionality and a low sample size, such as cancer classification-based microarray datasets. It is considered a challenge because of the huge number of gene subset which make the data unstable and prone to overfitting. Further, a smaller gene subset is appropriate to the classification procedure and the proportional overlapping result is a robust method because it delivers lower error rates [7].

Choosing the most appropriate methods of feature selection are essential during the preprocessing stage. Doing this will enhance the analysis and evaluation of the cancer diagnosis, prediction, and classification. The literature shows that there are many feature selection approaches that have been suggested by researchers to decrease the training time and computational complexity during designated the prediction models. Three main feature selection approaches are available: filter, wrapper, and embedded methods. These techniques were classified according to their associations with the classifiers [8]. The filter method means picking the features depending on the correlation between features independent from the classifier as in Ref. [9]. On the other hand, the wrapper technique assesses the effectiveness of the attributes by optimizing the accuracy during the model training. In contrast, in the embedded techniques, the feature selection process is associated with the classification phase as in Refs. [10,11]. In this approach, the process of selecting the most important set of features is integrated in the structure of the classification model to offer a measurement for selecting the features [12].

The dimensionality of the microarray is very high, thus expanding the dimensions of the search space. Therefore, an efficient search algorithm is required to shrink the dimensionality and overcome the overfitting problem that may occur during the classification procedure. Thus, in the current research we have suggested a new algorithm based on an adaptive memory to enhance the feature selection procedure: this is called 'memory-based cuckoo search algorithm'. The memory is built adaptively based on the solutions' fitness values and the improvement behavior (count of improvement) to keep the history of the previous solutions at each iteration. The support vector machine is used for building the classification model, according to its classification robustness, which can be proved by comparing its efficiencies with other algorithms [13].

2. Microarray datasets

A description of twelve microarray datasets is provided in this section. Table I shows a brief description of a published dataset employed in the current research. The colon cancer dataset (Alon 1999) proposed by Alon in 1999 consists of 2000 genes and 62 instances [14]. 40 instances were biopsies from tumors and 22 were normal instances of biopsies of healthy cells of the colons from the same patients. In a leukemia dataset (Golub 1999) presented by Golub et al., in 1999 [15], the dataset has 72 samples and 7129 genes. In a lung cancer dataset (Gordon 2002) proposed by Gordon, in 2002 [16], the dataset had 12533 genes and 181 tissue samples divided into two classes: 150 adenocarcinomas and 31 malignant pleural mesothelioma.

oBreast Cancer (Gravier2010), proposed by Gravier in 2010 [17], contained 2905 genes and 168 samples split into (Pomeroy 2002).
oTumor, proposed by Pomeroy in 2002 [18], contains 7128 genes and 60 samples split into two classes: 21 survivors and 39 failures.

- o Lymphoma (Shipp2002) dataset proposed by Shipp in 2002 [19], contains 6817 genes and 58 samples split into two classes. 32 from recovered patients and 26 from patients having fatal conditions.
- o Subramanian's (2005) dataset contains 10,100 genes and 50 samples split into two classes [20].
- o Myeloma (Tian 2003) proposed by Tian in 2003 [21], contains 12, 625 genes and 173 samples.
- o Breast Cancer (West 2001) dataset proposed by West in 2001 [22] contains 7129 genes and 49 sample splits into 2 classes.
- o Leukemia (Chiaretti 2004) dataset proposed by Chiaretti in 2004 [23], contains 12,625 genes and 128 samples distributed into 2 classes.
- o Glioma (Sun 2006) proposed by Sun in 2006 [24] dataset contains 54,613 genes and 180 samples split into 4 classes.
- o Leukemia (Yeoh 2002) proposed by Yeoh in 2002 [25] contains 12, 625 genes and 248 samples distributed into 6 classes.

3. The proposed cuckoo search feature selection algorithm

The Cuckoo Search (CS) algorithm was proposed in 2009 by Yang and Deb [26]. It is inspired from the obligate brood parasitic behavior of cuckoo leaves its egg in the nest of some other smaller bird, such as a starling. When the egg hatches the starling raises the cuckoo chick as if it were its own. As it begins to grow bigger than the other chicks in the nest, the cuckoo chick crowds out the others, to the point that they are forced out of the nest entirely. In CS algorithm the nest represents the population of the solutions, each egg signifies a possible solution in the nest and the cuckoo eggs signify the new solution(s). Proposed by Levy flight, a better new solution is replaced by a worse solution. The CS algorithm has three main rules which are described as follows [26]:

- Every cuckoo picks a nest at random and lays a single egg in it.
- The ideal nest with top quality eggs is ignored, to be improved in the following iteration.
- In each nest (predefined number of nests), the host cuckoo discovers a strange egg by probability ϵ [0,1]. Once the strange egg is discovered the host cuckoo decides either to abandon the egg or to leave the nest and create another nest in a different place.

CS algorithm discovers the new solutions by using the Markov chain model, in the CS algorithm the method is called a Levy flight equation (1) [26].

$$xi = xi + \alpha \oplus Levy(\lambda) \quad (1)$$

where xi is the solution, α is the step size where $\alpha > 0$, $\alpha = 1$ in most cases; λ is the Levy distribution co-efficient for a new solution for a cuckoo i.

The previous formula is the stochastic formula for random steps. Commonly, a random step is a Markov chain in which the following position is exclusively based on the present position (as 1st term in formula 1), and the 2nd term in formula 1 is the transition probability, $\oplus$ in the formula denotes the entry-wise multiplications. Where the step length is multiplied by randomly generated values with Levy's

```
Initial_population; initialize the population with Nst host nests;
Evaluate the Initial_population;
Set the max_iter;
Iter=0;
while(iter<max_iter)
  C= select_random_cuckoo; //select a random cuckoo
  C*= levy_flights(C);//apply levy_flights on C to generate new solution
  FC*=Evaluatefitness(C*); //compute the fitness of C*
  N=random_nest();//select a nest at random among Nst
  FN=Evaluatefitness(C); // compute the fitness of C
  if (FC* > FN)
    N = C*;// Replace N by the new generated solution C*;
  end if
  Abandon the worst pa nests;// where pa is a fraction of nests
  Construct new nests using levy_flights;
  save the best nests;
  Find the current best nests;
  iter=iter+1;
end while
```

Fig. 1. CS algorithm Pseudocode [26].

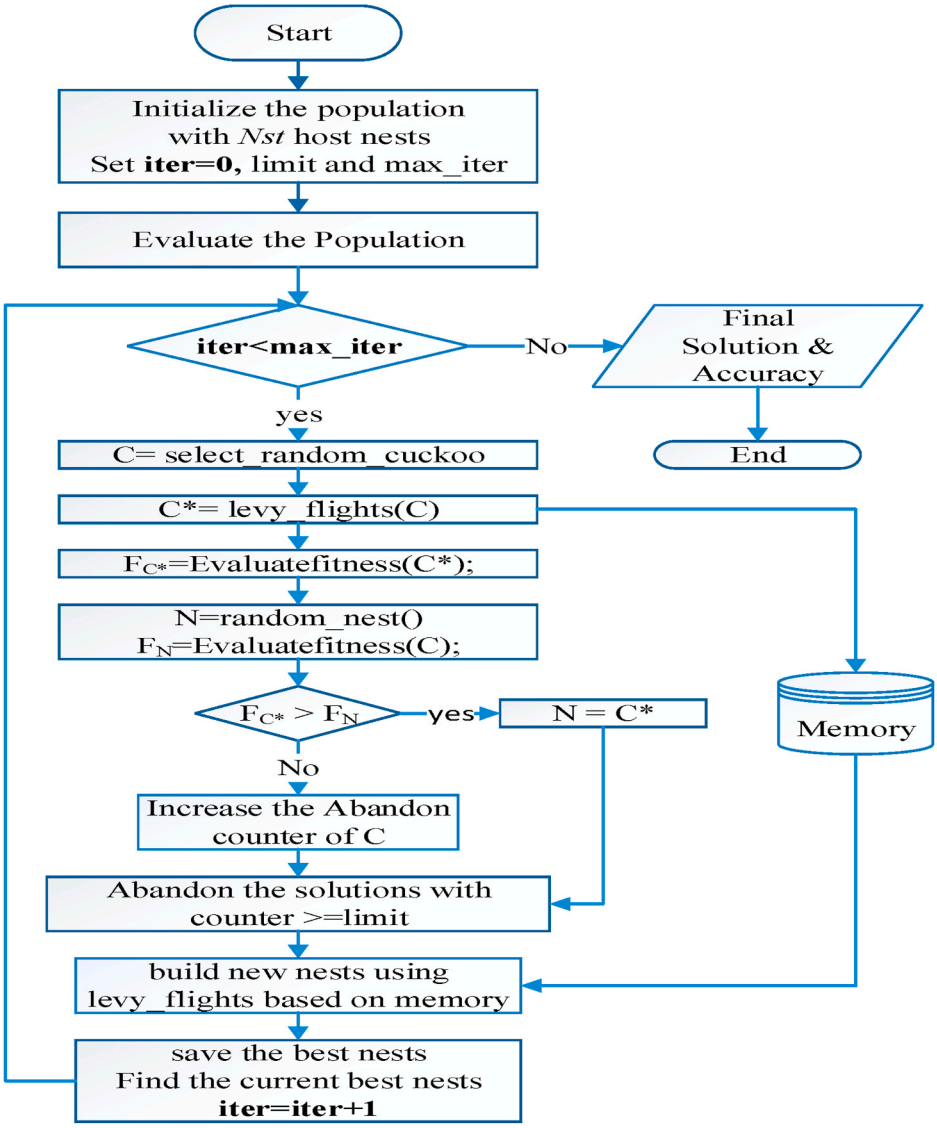


Fig. 2. The proposed Memory-Based Cuckoo Search Algorithm.

Table 2
CS algorithm Parameters.

Parameters	Value
population size	10
Iterations	100
Pa	0.3
A	1
λ	1.5

distribution, the random motion here is the Levy flight. In Levy flight, the random step works better in exploring the search space. Essentially, Levy flights afford a random step, while the random step sizes are given from a Levy distribution for big steps assumed in formula 2 [26].

Typically, the range of λ is in the range [1: 3]. The cuckoo's consecutive jumps here efficiently make a random steps procedure that responds to a power-law step-size distribution with a large tail. Yet, for the new solutions, a substantial fraction must be produced by far-field randomization and these positions would be sufficiently distant from the existing best solution. This will ensure that the system will not be stuck in a local optimum. The randomization here is further effective as the step length is large and heavy-tailed.

The CS algorithm starts with randomly initializing the population, then in each improvement iteration a random solution (cuckoo) is selected. A Levy flight is applied to the selected solution to generate a new solution, then a random location of the solution in the population (nest) is selected. The new solution will be located in a selected location if it is better than the solution in that location. Fig. 1 shows the whole process of the CS Algorithm.

In Fig. 1 fitness is the prediction accuracy, where the objective is to maximize the prediction accuracy (fitness) produced by the support vector machine (SVM). SVM is used to build the classification model, based on the classification robustness. This is tested by comparing its efficiencies with other algorithms [13].

4. Memory-based cuckoo search algorithm

The basic CS algorithm uses *Pa* as a fraction of the abandoned nests, and generates completely new solutions. This step seems not to be effective for the early stage of the process of CS algorithm since the solutions in the early stage have had no chance for further improvement. Further, at the late stage the solutions will lose some valuable information since they become abandoned, and it will be very hard to start a new solution to improve because no sufficient iterations are left. Thus, a simple memory is proposed to prevent abandoning the solutions in early or late stages. It will also keep the unimproved counter (the number of changes that didn't improve the solution) and the best neighbor solution found so far in the memory. This will help in choosing the abandoned solutions. Fig. 2 shows the process of the Memory-Based Cuckoo Search Algorithm.

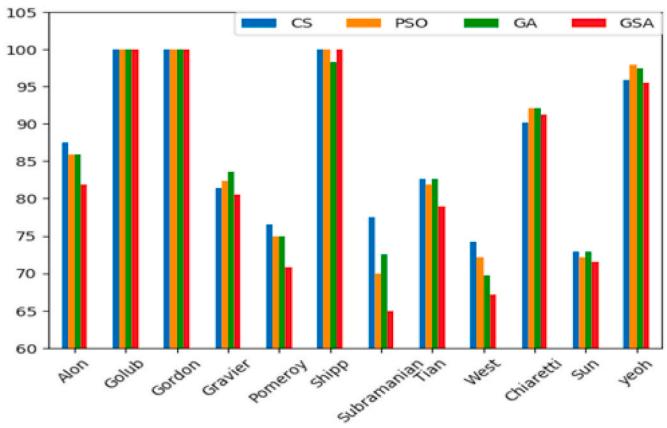


Fig. 3. Comparison between CS, PSO, GA and GSA algorithms in term of Acc.

The proposed algorithm consists of two main phases, as follow:

4.1. Building the memory

This phase is to keep some solutions with their previous state and fitness values are in the following matrix:

Sol 1-0	Sol 1-1	Sol 1-2	..	Sol 1-MS
Sol 2-0	Sol 2-1	Sol 2-2	..	Sol 2-MS
:	:	:	:	:
:	:	:	:	:
Sol N-0	Sol N-1	Sol N-2	..	Sol N-MS

Where the rows representing the solutions in population columns are the memory size (MS). A column with index *0* represents the best solution found so far. Since the memory is adaptive, the memory size is set to be the number of iterations divided by the population size. Therefore, if more iterations are performed a memory with bigger size is required.

4.2. Updating strategy

Once the improved solution found, then it will be pushed at index 1 (index 0 is used to keep the best solution), then the other solutions in the same row will be shifted to the right, removing the last one from the memory. This will offer to keep the last improved MS solutions so that, after number of non-improved moves, the algorithm will step back to the previous solution and try to find a new solution different from the one in hand.

5. Findings and discussion

The present section investigates the performance of the suggested

Table 3
Comparison between CS, PSO, GA and GSA in terms of accuracy and average accuracy.

Dataset	CS		PSO		GA		GSA	
	Acc	Avg-Acc	Acc	Avg-Acc	Acc	Avg-Acc	Acc	Avg-Acc
Alon	**87.5**	86.5	85.9	85.0	85.9	85.1	81.9	81.8
Golub	**100.0**	98.6	**100.0**	97.7	**100.0**	97.2	**100.0**	97.2
Gordon	**100.0**	100.0	**100.0**	100.0	**100.0**	100.0	**100.0**	100.0
Gravier	81.4	80.5	82.4	80.3	**83.6**	82.4	80.6	80.4
Pomeroy	**76.5**	74.5	75.0	72.2	75.0	75.0	70.8	70.8
Shipp	**100.0**	98.6	**100.0**	100.0	98.3	98.3	**100.0**	98.9
Subramanian	**77.5**	77.3	70.0	69.2	72.5	68.3	65.0	64.2
Tian	**82.6**	80.1	81.9	75.8	**82.6**	81.8	78.9	78.4
West	**74.2**	65.7	72.2	70.7	69.7	67.2	67.2	66.4
Chiaretti	90.2	89.8	**92.1**	83.9	**92.1**	89.3	91.2	88.9
Sun	**72.9**	71.1	72.2	71.9	**72.9**	72.0	71.5	71.0
yeoh	95.9	95.7	**97.9**	97.6	97.5	96.6	95.46	95.1

Table 4
Comparison of the CS with two versions using Accuracy and Selected features metrics.

Dataset	CS					Modified CS				
	Acc	Avg Acc	Std. Dev	SF	Avg SF	Acc	Avg Acc	Std. Dev	SF	Avg SF
Alon	87.5	86.5	0.0112	1311	1293.71	**94.0**	90.6	0.0193	201	202.6
Golub	**100.0**	98.6	0.0067	1862	4702	**100.0**	100.0	0.0000	650	715.6
Gordon	**100.0**	100.0	0.0000	8146	6187.5	**100.0**	100.0	0.0000	1206	1262.5
Gravier	81.4	80.5	0.0073	1883	1864.3	**84.5**	82.9	0.0097	299	291.9
Pomeroy	76.5	74.5	0.0187	**251**	3921.7	**86.0**	81.3	0.0285	754	713.2
Shipp	**100.0**	98.6	0.0057	4642	4236.1	**100.0**	100.0	0.0026	655	712.3
Subramanian	77.5	77.3	0.0062	6588	5794.5	**87.5**	81.1	0.0201	998	1009.4
Tian	82.6	80.1	0.0078	8141	8185.2	**84.1**	82.1	0.0088	1287	1259.1
West	74.2	65.7	0.0243	**425**	3691.2	**81.7**	78.4	0.0216	684	714.6
Chiaretti	90.2	89.8	0.0048	8105	8150.3	**92.1**	91.1	0.0087	1221	1268.7
Sun	72.9	71.1	0.0097	16375	26145.8	**77.8**	75.3	0.0215	5516	5457.4
yeoh	95.9	95.7	0.0025	3464	3636.5	**98.0**	97.4	0.0041	1244	1263.6

method when tested in twelve microarray datasets, the description of which was presented in Table 1. The parameter settings of the CS algorithm, which have been chosen based on primary experiments, are listed in Table 2. In the basic CS algorithm, the *pa* and α parameters are fixed [26]. The parameters *pa*, α, and λ in the CS algorithm are respectively set to 0.3, 1 and 1.5, where the memory size is set to be the number of iterations divided by the population size.

All the results used for comparison in this paper were performed on a PC with Intel i5-6200 2.30 GHz CPU and 8.0 GB RAM. Also, the statistical results were conducted over 31 separate runs. The algorithms were compared in terms of number of features and classification accuracy. The datasets are divided into testing and training groups, where 80% of the microarray dataset were used in the training phase and the remaining 20% was used for testing purposes [27].

This section represents the evaluation of the results of four well-known population-based techniques (the CS algorithm, the Gravitational Search Algorithm (GSA), the genetic algorithm (GA) and Particle swarm optimization (PSO)) which were investigated to judge the performance of the proposed method. The evaluation of the offered methods is done in terms of the accuracy and the average classification accuracy (Acc). Table 3 shows the accuracy (Acc) and Average Accuracy (Av-Acc) for CS, PSO, GA, GSA algorithms. The values with bold fonts represent the best among other values. It can be observed that the CS algorithm is able to reasonably surpass other competitors with regards to Acc and Average Acc, in dealing with twelve datasets.

Fig. 3 illustrates the accuracies and average accuracies found by the CS algorithm and three population-based algorithms (PSO, GA and GSA) of all the microarray datasets used in this work. We can observe that the algorithms –CS, PSO, and GSA-achieve 100% of the classification accuracy in three datasets (Golub, Gordon and Shipp), while GA algorithm obtained 100% accuracy in two datasets (Golub and Gordon). In general, The CS algorithm can reveal higher accuracies than the other variant in 9 out of 12 datasets.

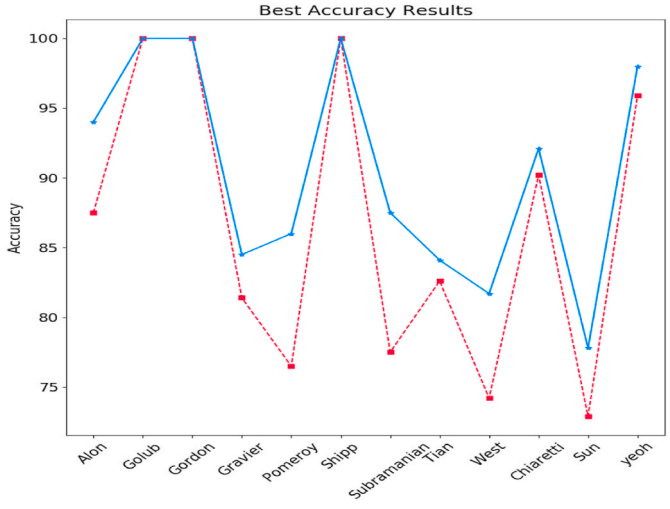


Fig. 4. Comparison of the CS and Modified CS algorithms in term of Accuracy.

5.1. Evaluation of the proposed method

In this section, the outcomes of the two algorithms in comparison (the CS algorithm and memory-based CS (Modified CS)) are investigated to judge the performance of the suggested method. The evaluation of the suggested methods is done with respect to the number of picked features, the average number of picked features, overall accuracy, average accuracy, and the P-values of the *t*-Test of each technique.

Table 4 shows the accuracy, average accuracy, standard deviation, selected feature, and average selected features for the CS algorithm versus the modified version for the algorithm. Moreover, it is noticed that the Modified CS can relatively surpass other competitor's metrics in dealing with twelve datasets. The CS algorithm is unable to present higher accuracies as compared to other variants over all twelve datasets. For the Golub, Gordon and Shipp datasets, the CS algorithm and the Modified CS algorithm obtained 100% of classification accuracy; nevertheless, to achieve this accuracy the CS algorithm used 1862, 8146 and 4642 features in Golub, Gordon and Shipp datasets, respectively, while Modified CS used only 650, 1206 and 655 features.

The Modified CS algorithm attained 100% in solving the Alon, Golub, Gordon, and Shipp datasets. Table 4 shows that the selected features and average selected features can provide inferior numbers when contrasted to the original CS algorithm in all considered datasets. Based on the selected features in Table 4, it is observed that the modified

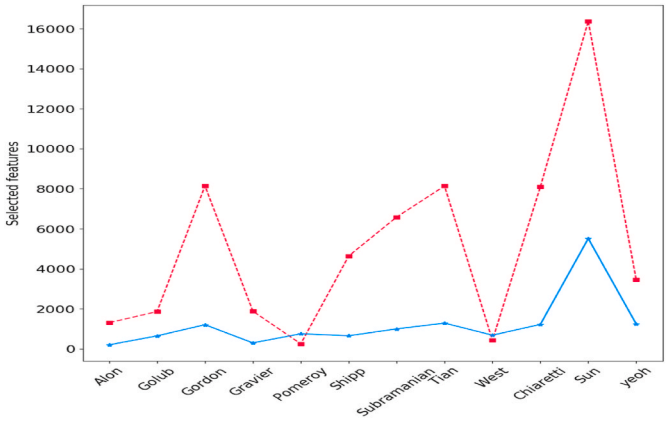


Fig. 5. Comparison between the CS and Modified CS algorithms in terms of selected features for all datasets.

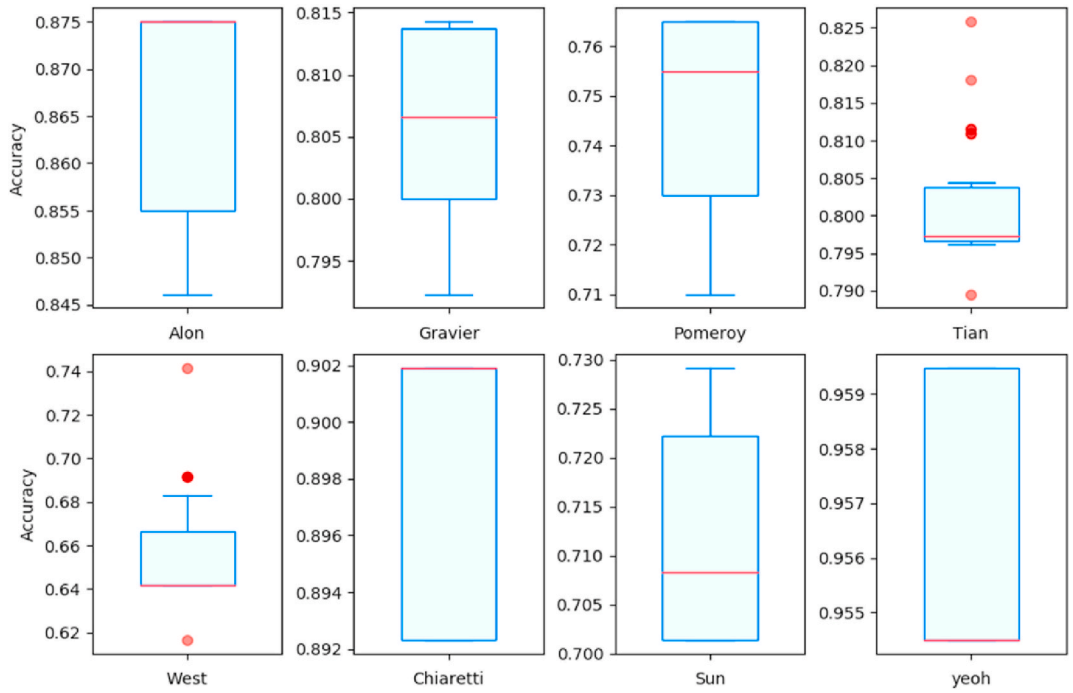


Fig. 6. Box-and-whisker plots for the microarray datasets tested on the CS algorithm.

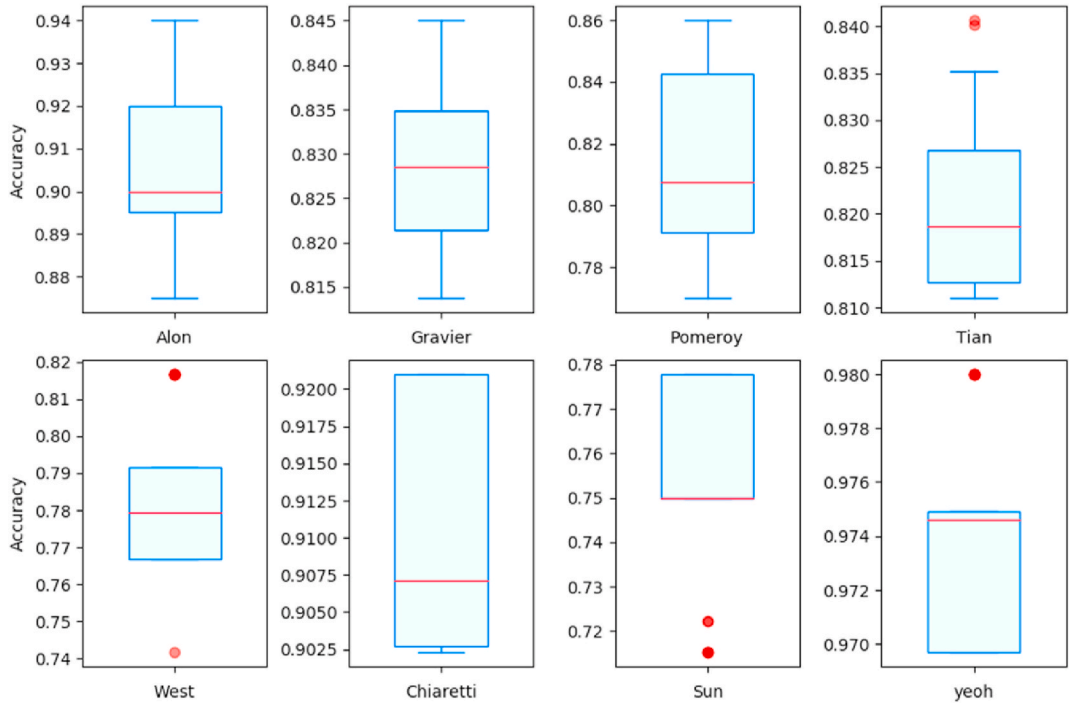


Fig. 7. Box-and-whisker plots for the microarray datasets tested on the Modified CS algorithm.

CS algorithm significantly decreases the number of features selected, comparing to the original dataset numbers. Also, the Modified CS is better than the original on 10 datasets (except for **Pomeroy** and **West** datasets) in terms of selected features. This illustrates that the modification affects the results that, because of the algorithm, keep the solutions visited and memorize some valuable information. This helps in further investigations and searches as it will prevent abandon the solutions in early or late stages. Then it is obviously observed that the Modified CS is superior when compared with the CS algorithm on all comparisons for twelve datasets.

In Figs. 4 and 5, each line represents the number of picked features (#SF), the Accuracy value for the CS algorithm, and the modified version of the CS algorithm for each Microarray dataset, respectively. The upper line in Fig. 4 denotes the Modified CS algorithm, and the upper line in Fig. 5 denotes the CS algorithm.

Figs. 6 and 7 provide the box and whisker plot, which displays the spread of the accuracy values for each solution obtained from CS and Modified CS algorithms for eight datasets, showing the distribution of

Table 5
Comparison of the P-value with two versions using accuracy, average accuracy, and selected features metrics.

Dataset	P-Value between		
	Best Accuracy. CS Vs. Modified CS	Average Accuracy. CS Vs. Modified CS	#of Selected features. CS Vs. Modified CS
Alon	**3.74E-14**	0.32	**9.35E-51**
Golub	**1.19E-11**	0.08	**1.95E-14**
Gordon	*NA*	**0.005**	**1.32E-12**
Gravier	**2.51E-15**	**5.1E-05**	**2.00E-46**
Pomeroy	**1.18E-19**	0.45	**1.82E-12**
Shipp	**3.75E-13**	**7.17E-05**	**8.94E-18**
Subramanian	**6.32E-11**	0.9000	**8.02E-15**
Tian	**3.58E-10**	**0.0300**	**6.17E-60**
West	**8.36E-18**	0.4200	**7.61E-11**
Chiaretti	0.0987	**1.00E-5**	**0.0424**
Sun	0.0660	**1.00E-5**	0.2950
Yeoh	**0.0005**	**1.00E-5**	**0.0333**

Table 6
Comparison between the Modified CS algorithm with state-of-the-art algorithms.

	Modified CS algorithm	The state-of-the-art Acc results	
Dataset	Acc	The State-of-the-art	Acc
Alon	**94.0**	Huijuan et al., 2017 [11]	83.41
		Mundra et al., 2010 [31]	91.10
		Mafarja et al., 2017 [33]	87.00
Golub	**100.0**	Huynh et al., 2018 [29]	**100.00**
		Cilia et al., 2019 [30]	99.44
		Güçkiran et al., 2019 [28]	**100.00**
Gordon	**100.0**	Huynh et al., 2018 [29]	**100.00**
		Mundra et al., 2010 [31]	99.90
		Cilia et al., 2019 [30]	**100.00**
Gravier	**84.5**	Huynh et al., 2018 [29]	79.80
		Arias-Michele al 2015 [34]	76.20
		Güçkiran et al., 2019 [28]	79.80
Pomeroy	86.0	Mundra et al., 2010 [31]	**92.10**
		Güçkiran et al., 2019 [28]	85.00
		Arias-Michele al 2015 [34]	63.30
Shipp	**100.0**	Arias-Michele al 2015 [34]	92.30
		Mundra et al., 2010 [31]	92.10
		Güçkiran et al., 2019 [28]	**100.00**
Subramanian	**87.5**	Güçkiran et al., 2019 [28]	**96.00**
Tian	**84.1**	Huertas et al., 2016 [35]	78.70
		Güçkiran et al., 2019 [28]	82.70
West	81.7	Güçkiran et al., 2019 [28]	**89.80**
Chiaretti	**92.1**	Huynh et al., 2018 [29]	85.20
		Güçkiran et al., 2019 [28]	88.30
Sun	77.8	Le Thi et al., 2017 [36]	72.30
		Güçkiran et al., 2019 [28]	**83.30**
Yeoh	98.0	Trevino & Martinez 2012 [37]	98.30
		Wang et al., 2007 [32]	**99.70**

our results. The median in each box plot is represented by the line dividing the box into two horizontal parts, the normal distribution is observed if the results are proportioned normally around the median.

From Fig. 6 we can observe that some data sets' results are proportioned normally around the median; like Gravier, Pomeroy and Sun. Data above the median is more dispersed. We can also notice five outliers at the higher extreme. But in Fig. 6 for the Modified CS algorithm, the observations for the 7 out of 8 datasets show symmetrical distributions with the minimum, median, and maximum being close to each other. Thus, the robustness of our Modified CS algorithm is confirmed.

5.2. Statistical test

The statistical significance test is used to determine if there is any major distinction between the two methods. The *t*-Test is employed in this work for testing the null hypothesis which means the results produced by the two algorithms are equal.

The above-described *t*-Test has been applied to the accuracy, average accuracy and the number of picked features of each microarray dataset from every one of the two algorithms available in Table 3. The null hypothesis of the *t*-Test considered here is that there is no difference between the two versions of CS Algorithm based on accuracies and the number of features.

The decision rule then says that we must reject the null hypothesis when the P-value< 0.05. While the P-value $= 0.417 > 0.05$, then the null hypothesis is accepted. We may then decide that there is no difference between the CS Algorithm and the Modified CS Algorithm in terms of accuracy.

As well, if P-value $= 0.013 < 0.05$, the null hypothesis is rejected, and we decide that there is a difference between the CS Algorithm and the Modified CS Algorithm in terms of average accuracy. With similar calculations, the *t*-Test has been applied to selected features and average selected features of each microarray dataset from each of the two different algorithms.

The null hypothesis of the *t*-Test is that there is no difference in the mean between the two CS Algorithms based on selected features. Respectively, if the P-value $= 0.017 < 0.05$, and P-value $= 0.0002 <$ 0.05 then the null hypothesis is rejected and we decide that there is a significant difference between the CS Algorithm and the Modified CS Algorithm in terms of selected features and average selected features. It was found that the accuracy and the number of selected features in the Modified CS Algorithm are statistically better than those of the CS Algorithm. Considering all results; accuracies, and selected features of the two algorithms, it can be recognized that the Modified CS Algorithm is superior. It can generate the highest accuracy with a minimum number of features when applied to some of the cancer microarray datasets.

Table 5 describes the result of P-value applied on the accuracy, average accuracy, and selected features of each microarray dataset from each of the two different algorithms. Note that the fields with *NA* in Table 5 mean that the results for the 31 runs are the same.

The P-values with bold fonts in Table 5 are significant (P-value$<$ 0.05). The decision rule then says that we should reject the null hypothesis if P-value< 0.05 and we will decide that there is a difference between the CS Algorithm and the modified CS Algorithm in terms of accuracies, average accuracy, and selected features.

Since the P-values with plain-font text are not significant (P-value$>$ 0.05), the decision rule then says that we should accept the null hypothesis if P-value > 0.05 and we will decide that there is no difference between the CS Algorithm and the Modified CS Algorithm in terms of accuracies, average accuracy, and selected features. There are four datasets in the comparison that are significant for all metrics. The p-value is < 0.05 in 9, 7, and 11 datasets in terms of accuracy, average accuracy, and number of feature selection respectively. Considering all results, the modified CS Algorithm can generate the highest accuracy with a minimum number of features when applied to some of the cancer microarray datasets.

Table 6 shows the comparison between the Modified CS algorithm with the state-of-the-art algorithms depending on the best accuracy obtained from these algorithms, where the bold font illustrates the best accuracy. Table 6 show that the Modified CS algorithm achieves 8 best results out of 12 datasets, Güçkiran et al. [28], get 5 best results, Huynh et al. [29], achieve 2, followed by Cilia et al., 2019 [30], Mundra et al. [31], Wang et al. [32], achieving one best result. Thus, the Modified CS algorithm is superior compared with all other algorithms.

6. Conclusion

In this paper, a feature selection algorithm guided by the memory-based mechanism was proposed and tested on high-dimensional microarray datasets. The proposed two versions of the CS algorithm (CS and Modified CS) were used extensively to tackle twelve microarray

datasets. The modified CS uses the memory-based mechanism, which helps the algorithm to memorize the solution behavior. The features caused the improvements during the search process and kept a count of each solution in the population. This saves the number of iterations and improves each solution in the population. The overall classification accuracy, selected features and average accuracy of two versions were compared in detail. The modified CS algorithm has obtained the best place among the original hybrid CS Algorithm for feature selection. Furthermore, when the Modified CS algorithm was compared to the state-of-the-art algorithms it was comparable and superior. Evaluation of the effectiveness of proposed memory mechanisms in other population-based algorithms and different problems such as vehicle routing problem and flexible job shop scheduling will be our subject of future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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